MODELS AND DIALOUGES

LEARNING OBJECTIVES

At the end of the session, students are expected to:

- Analyze and identify user (cognitive) models, socio-organizational issues, and stakeholder requirements in HCI systems
- Analyze and formulate tasks and dialogues using task analysis and dialogue design

COGNITIVE MODELS

Cognitive models attempt to represent the internal mental processes of users while interacting with a system. These include:

- Perception (what users see)
- Memory (what they remember)
- Attention (what they focus on)
- Decision-making (what action they take)

In HCI, we are not just designing interfaces—we are designing for the human brain.

TYPES OF COGNITIVE MODELS

Mental Model

- User’s understanding of how a system works
- Example: Users expect a trash icon = delete

GOMS Model (Goals, Operators, Methods, Selection Rules)

- Breaks down user interaction into steps
- Used to measure efficiency

Keystroke-Level Model (KLM)

- Predicts how long tasks take

SOCIO-ORGANIZATIONAL ISSUES & STAKEHOLDER REQUIREMENTS

HCI systems don’t exist in isolation—they are affected by:

- Organization
- Culture
- People involved

SOCIO-ORGANIZATIONAL ISSUES

Factors affecting system use:

- Work environment
- Communication flow
- User roles
- Cultural differences

STAKEHOLDERS

People affected by the system:

- Primary users (direct users)
- Secondary users (indirect users)
- Tertiary stakeholders (management, regulators)

ANALYZING TASKS

Task analysis is the process of:

- Breaking down user activities into steps

TYPES OF TASK ANALYSIS

Hierarchical Task Analysis (HTA)

- Break tasks into subtasks

Use Case Analysis

- Focus on user-system interaction

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TYPES OF TASK ANALYSIS

Example

Task: “Login to system”

Steps:

1. Open app
2. Enter username
3. Enter password
4. Click login

DIALOGUE NOTATION AND DESIGN

Dialogue design defines:

- How users and systems communicate

TYPES OF DIALOGUE STYLES

1. Command Line
2. Menu Selection
3. Form Fill-in
4. Direct Manipulation (drag/drop)
5. Natural Language (chatbots)

DIALOGUE NOTATION

1. Used to represent interaction flow:
2. State Transition Diagrams
3. Flowcharts

EXAMPLE

ATM interaction:

1. Insert card
2. Enter PIN
3. Select transaction
4. Confirm

DESIGN PRINCIPLES

1. Consistency
2. Feedback
3. Error handling
4. Simplicity

THANK YOU